

Solutions to *Group Theory in a Nutshell for Physicists*

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Abstract

This document provides the solutions I have worked out from the book *Group Theory in a Nutshell for Physicists* by Zee [1].

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PART I

I.1 Symmetry and Groups

Problem 1:

The center of a group G (denoted by Z) is defined to be the set of elements $\{z_1, z_2, \dots\}$ that commute with all elements of G , that is, $z_i g = g z_i$ for all g . Show that Z is an abelian subgroup of G .

Solution:

Let's start by showing that $\{z_1, z_2, \dots\} = Z$ form a group. The elements $\{z_1, z_2, \dots\}$ inherit associativity from being in the group G . Since the elements $\{z_1, z_2, \dots\}$ are in the center of G , they include the identity element. The inverse of the elements $\{z_1, z_2, \dots\}$ are also in the group:

$$\begin{aligned} z_i g &= g z_i \\ z_i^{-1} z_i g z_i^{-1} &= z_i^{-1} g z_i z_i^{-1} \\ g z_i^{-1} &= z_i^{-1} g, \end{aligned}$$

which means $z_i^{-1} \in Z$. The multiplication of elements of Z are in Z :

$$\begin{aligned} z_k &= z_i z_j \\ g z_k &= g z_i z_j \\ g z_k &= z_i z_j g \\ \Rightarrow g z_k &= z_k g \end{aligned}$$

so $z_k = z_i z_j \in Z$ for all i and j . So Z satisfies all the properties of being a group, and since it's made up of elements in G , it is a subgroup of G . To show Z is abelian is trivial from the definition of center:

$$z_i g = g z_i \quad \forall g \in G \quad \Rightarrow \quad z_i z_j = z_j z_i \quad \forall z_i, z_j \in Z,$$

since $Z \subset G$, meaning Z is abelian subgroup of G .

Problem 2:

Let $f(g)$ be a function of the elements in a finite group G , and consider the sum $\sum_{g \in G} f(g)$. Prove the identity $\sum_{g \in G} f(g) = \sum_{g \in G} f(g'g)$ for g' an arbitrary element of G . We will need this identity again and again in chapters II.1 and II.2.

Solution:

For a finite group G , $g' \cdot g_i$ is unique for all $g_i \in G$ for arbitrary $g' \in G$. So, the elements $g' \cdot g_i$ form a group which is the original group G that we started with. This means when we sum over the index i , we sum over all the elements of G , no matter how many arbitrary elements that multiply g_i . This means $g'g$ is equivalent to renaming the elements, so we can write $\sum_{g \in G} f(g'g) = \sum_{k \in G} f(k)$. The same thing applies to $g_i \cdot g'$, because both $g' \cdot g$ and $g \cdot g'$ are mappings $G \rightarrow G$. Using this we get

$$\sum_{k \in G} f(k) = \sum_{g \in G} f(gg') = \sum_{g \in G} f(g'g),$$

which is the identity that was to be proved. (Here we renamed to k , but it can be g too, obviously.)

Problem 3:

Show that $Z_2 \otimes Z_4 \neq Z_8$.

Solution:

The groups can be defined by

$$Z_2 = \{I, A\}, \text{ with } A^2 = I$$

$$Z_4 = \{I, A, A^2, A^3\}, \text{ with } A^4 = I$$

$$Z_8 = \{I, A, A^2, \dots, A^7\}, \text{ with } A^8 = I$$

The group $Z_2 \otimes Z_4$ can be constructed using the composition

$$(n, m) \cdot (n', m') = (n + n' \pmod 2, m + m' \pmod 4)$$

Let's find all the 8 elements of this group starting from the identity $I = (0, 0)$ and multiplying by elements $+(1, 1)$. This gives us half of the elements while the other half is gotten by starting from $(0, 1)$ and multiplying by elements $(1, 1)$. The four elements from multiplying identity with $(1, 1)$ are

$$(0, 0) \xrightarrow{+(1,1)} (1, 1) \xrightarrow{+(1,1)} (0, 2) \xrightarrow{+(1,1)} (1, 3) \xrightarrow{+(1,1)} (0, 0).$$

The other four elements are

$$(0, 1) \xrightarrow{+(1,1)} (1, 2) \xrightarrow{+(1,1)} (0, 3) \xrightarrow{+(1,1)} (1, 0) \xrightarrow{+(1,1)} (0, 1).$$

From this we can see that the group $Z_2 \otimes Z_4$ has the structure

$$Z_2 \otimes Z_4 = \langle A, B | A^2 = I, B^4 = I, AB = BA \rangle.$$

Meanwhile, the group structure of Z_8 is

$$Z_8 = \langle A | A^8 = I \rangle.$$

Comparing the two group structures we can clearly see that

$$Z_2 \otimes Z_4 \neq Z_8.$$

Problem 4:

Find all groups of order 6.

Solution:

(This solution is a bit lengthy and messy, since I want to show my thought process on calculating the Cayley tables with most of the steps being shown.)

One group of order 6 is Z_6 . To find other groups of order 6, we can construct Cayley table that is not the table of Z_6 .

NOTE: Lagrange theorem: Let a group G with n elements have a subgroup H with m elements. Then m is a factor of n

This means when we have elements $G = \{I, A, B, C, D, E\}$ we can only have $A^6 = I$, which fixes the group to be Z_6 , or $A^3 = I$ or $A^2 = I$ along with same considerations for other elements.

Let's start by constructing a Cayley table that is not of Z_6 .

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A					
B	B					
C	C					
D	D					
E	E					

For a system with $A^3 = I$ we would need to have $B = A^2$. (Substituting other letter at this point for B is akin to renaming, so it's not important to look what $C = A^2$ looks like.)

this results in

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	B	I			
B	B	I	A			
C	C					
D	D					
E	E					

Now, we cannot have $CA = I$ or $CB = I$. So let's consider $C^2 = I$, which leads to $C^2 = A^3 = BA$, so the only option is to have $CA = D$, $DA = CB = E$, $EA = DB = C$. This gives $CD = A$, $CE = B$, $DD = B$, $EE = A$, and $DE = I$. This gives the Cayley table

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	B	I	D	E	C
B	B	I	A	E	C	D
C	C	D	E	I	A	B
D	D	E	C	A	B	I
E	E	C	D	B	I	A

This is group $G = \langle A, C | A^3 = I, C^2 = I \rangle$ which is just $Z_2 \otimes Z_3$, and it is isomorphic to Z_6 , so this is not a new group of order 6.

Now the only scenario left is one where $A^2 = I$

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	I				
B	B					
C	C					
D	D					
E	E					

Once again we cannot have $B^3 = I$ as an option since it would lead to us creating a group isomorphic to Z_6 , so we must have either $B^2 = I$ or $BC = I$. Both of these choices actually lead to same result, so let's choose the first one. (We will see as we fill the Cayley table why they lead to same result.) We come across this same choice for C with D , so let's just choose $C^2 = I$. We then get

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	I	AB	AC	AD	AE
B	B	BA	I	BC	BD	BE
C	C	CA	CB	I	CD	CE
D	D					
E	E					

$AB = C \Rightarrow AC = B$ cannot work, because then we get the following table which is invalid as a Cayley table,

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	I	C	B	E	D
B	B	C	I	A	BD	BE
C	C	B	A	I	CD	CE
D	D	E	DB	DC	I	A
E	E	D	EB	EC	A	I

This is invalid because we don't have anywhere to place D and E in the B and C , column. Note that having $DE = I$ and $DD = A = EE$ wouldn't have solved the problem, since it's equivalent to renaming D and E (or swapping their rows or columns). This means that once and only once rule leaves us $AB = D$ and $AC = E$ or $AB = E$ and $AC = D$. Once again both of these choices are equivalent since it's just renaming of labels. So, let's choose one where $AB = D$, which implies then $AD = B$, and $AE = C$. By similar argument we cannot have B or C in BA and CA , which leaves us with $BA = E \Rightarrow EA = B$ and $CA = D \Rightarrow DA = C$. Our Cayley table then becomes

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	I	D	E	B	C
B	B	E	I	BC	BD	BE
C	C	D	CB	I	CD	CE
D	D	C	DB	DC	DD	DE
E	E	B	EB	EC	ED	EE

Here, the choice of $DE = I$ or $DD = I$, matters. Of these only the first one leads to a valid Cayley table. Using $BA = E$ and $AC = E$ we get $BAAC = BC = EE$. Similarly $CA = D$ and $AB = D$ which gives us $CB = DD$. Our Cayley table then becomes

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	I	D	E	B	C
B	B	E	I	EE	BD	BE
C	C	D	DD	I	CD	CE
D	D	C	DB	DC	DD	I
E	E	B	EB	EC	I	EE

Once and only once rule means we need $EB = C$, $BD = C$, $CD = E$, $DD = A$

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	I	D	E	B	C
B	B	E	I	EE	C	BE
C	C	D	A	I	E	CE
D	D	C	DB	DC	A	I
E	E	B	C	EC	I	EE

Once and only once rule means $DC = B$, $DB = E$, $CE = B$

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	I	D	E	B	C
B	B	E	I	EE	C	BE
C	C	D	A	I	E	B
D	D	C	E	B	A	I
E	E	B	C	EC	I	EE

Since $BC = EE$ we can have both $BC = D = EE$ and $BC = A = EE$, but this amounts to just renaming $C \rightleftharpoons E$. So let's choose $BC = EE = A$, which gives us the only valid Cayley table that is not for Z_6 ,

	I	A	B	C	D	E
I	I	A	B	C	D	E
A	A	I	D	E	B	C
B	B	E	I	A	C	D
C	C	D	A	I	E	B
D	D	C	E	B	A	I
E	E	B	C	D	I	A

This Cayley table is the only Cayley table of order 6 that is not isomorphic to Z_6 , and it is the Cayley table for the permutation group S_3 . This means that the only groups of order 6 (up to isomorphism) are Z_6 and S_3 .

Part II

Nothing here, the solutions are available in my website as handwritten solutions. There are solutions to exercises from Part III, so continue on.

Part III

III.1 Quantum Mechanics and Group Theory: Parity, Bloch's Theorem, and the Brillouin Zone

Problem 1:

Let ψ^a be a set of degenerate wave functions that transform under a symmetry group like $\psi^a \rightarrow \psi'^a = (D(T))^{ab}\psi^b$. Show that $D(T_2T_1) = D(T_2)D(T_1)$.

Solution:

We first apply the transformation $\psi^a \rightarrow \psi'^a = (D(T))^{ab}\psi^b$ on ψ'^a get the transformation $\psi'^a \rightarrow \psi''^a$.

$$\begin{aligned}\psi'^a \rightarrow \psi''^a &= (D(T_2))^{ab}\psi'^b \\ &= (D(T_2))^{ab}(D(T_1))^{bd}\psi^d \\ &= (D(T_2)D(T_1))^{ad}\psi^d\end{aligned}$$

This is one way to get $\psi^a \rightarrow \psi''^a$, but there is also another way: we apply both the transformations T_1 and T_2 on ψ^a directly

$$\psi^a \rightarrow \psi''^a = (D(T_1T_2))^{ad}\psi^d.$$

Comparing these two equations directly, we get

$$D(T_2)D(T_1) = D(T_2T_1) \quad \square$$

III.2 Group Theory and Harmonic Motion: Zero Modes

Problem 1:

Let us make the “ridiculously simple” example of two masses connected by a spring and moving in one dimension a bit less simple by having two unequal masses. Without writing down any equations, do you think that there is a zero mode?

Solution:

Yes, because there should be a mode where only the center of mass is moving. So, $\omega = 0$ just like in the equal mass case.

(Haven't actually checked the correct answer or done the calculation yet.)

Problem 2:

Use the Great Orthogonality theorem (II.2.2) to derive the projection operators into the desired modes.

Haven't done this (yet?).

References

- [1] A. Zee. *Group Theory in a Nutshell for Physicists*. In a Nutshell. Princeton University Press, 2016.